

FreshFlow Beverages Pvt. Ltd.

FreshFlow Bottling Plant - Pune Production Facility
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STANDARD OPERATING PROCEDURE

Cleaning Procedure for Utilities Line 18

Department: Utilities

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Approved by: Plant Manager

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Title	Cleaning Procedure for Utilities Line 18
Department	Utilities
Plant	FreshFlow Bottling Plant - Pune Production Facility
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1. Purpose

Safe operation of plant utilities (compressed air, steam, chilled water, CO2, electrical) to support continuous production.

2. Scope

Utilities department operators, Maintenance engineers.

3. Prerequisites and Requirements

3.1 Training Requirements

- Boiler operator certification (IBR)
- Compressor operation training
- Electrical safety training (LV)

3.2 Tools and Materials Required

- Calibrated instruments as listed in the plant calibration schedule.
- Appropriate PPE as specified in Section 4.
- Completed permit-to-work (if required for maintenance activities).
- Access to relevant equipment manual and P&ID drawings.
- CMMS access to create and close work orders.

4. Health, Safety and Environmental Requirements

! WARNING

This SOP must not be carried out without appropriate PPE and, where required, a valid Permit to Work (SOP-SAF-006). If in doubt about safety, STOP and consult the Safety Officer (EMP009 / EMP017).

4.1 Identified Hazards

- High-pressure steam (7 kg/cm²) - scald and explosion risk
- High-voltage electrical (415V / 11kV) - electrocution risk
- CO₂ stored at high pressure (-28°C liquid) - cryogenic and pressure hazard
- Diesel generator - hot exhaust, rotating parts, high voltage output

4.2 Emergency Response

In case of any emergency during this procedure: Press the nearest Emergency Stop, alert personnel in the vicinity, call the Safety Officer and Shift Supervisor. Refer to the Emergency Response Plan (SOP-SAF-004) and site evacuation procedure.

5. Detailed Procedure

5.1 Pre-Task Verification

1. Confirm the relevant equipment is available and no conflicting activities are planned.
2. Check shift handover log for any outstanding issues with this equipment.
3. Verify all required materials, tools, and chemicals are available and correctly labelled.
4. Obtain all required permits (PTW, hot work, confined space as applicable).
5. Brief all involved personnel on this SOP before starting.

5.2 Main Procedure

1. Confirm production has stopped and product has been drained from the system.
2. Ensure CIP supply pump P301 is ready and CIP tanks have correct chemical charge.
3. Connect CIP return line and verify all CIP circuit valves are in correct position.
4. Start pre-rinse with RO water for 5 minutes - verify runoff is clear.
5. Switch to caustic recirculation: 2% NaOH at 75°C for 20 minutes minimum.
6. Monitor caustic concentration with inline conductivity meter - maintain 15-20 mS/cm.
7. Flush with RO water until conductivity <100 uS/cm and pH 6.5-7.5.
8. Perform acid circulation: 1% HNO₃ at 65°C for 15 minutes (passivation step).
9. Final rinse with RO water until conductivity <20 uS/cm and pH 6.5-7.5.
10. Take microbiological swab from defined sample points. Record in QC log.
11. Equipment is released for production only after QC verification (Form FFB-QA-CIP-VER).